INPUT

ARCHITECTURAL INPUT

The design brief, expressed as rated architectural criteria.

TYPOLOGY
ARCHETYPE
CRITERIA
RATINGS
DESCRIPTORS

> Ratings and descriptors are locked. They are the brief every later stage reads.

CRITERIA RATINGS + DESCRIPTORS →

STAGE

PHYSARUM LOGIC

Architectural conditions become biological behaviour.

ARCHITECTURAL CONDITIONS
↓
BIOLOGICAL TRANSLATION
↓
AGENT BEHAVIOUR
↓
CANONICAL RECIPE
↓
2D MORPHOLOGY GENERATION

> The translation stays fixed. Its generated morphologies are what 2D evolution searches.

2D MORPHOLOGY + TRANSLATION STATE →

STAGE

2D EVOLUTION

Populations of morphologies are measured and searched against the criteria.

POPULATION GENERATION
↓
MORPHOLOGICAL MEASUREMENT
↓
FORMAL / SPATIAL / ATMOSPHERIC CORRESPONDENCE
↓
EVOLUTIONARY SEARCH
↓
PARETO ARCHIVE
↓
DESIGNER SELECTION

> The search returns non-dominated alternatives. The designer chooses which morphology proceeds.

SELECTED NON-DOMINATED MORPHOLOGY →

STAGE

VERTICAL PROPAGATION

The chosen section grows through successive states into volume.

SELECTED 2D MORPHOLOGY
↓
SUCCESSIVE STATES
↓
VERTICAL PROPAGATION
↓
VOLUMETRIC MORPHOLOGY

> The chosen section grows through successive states into one volumetric morphology.

15 VOLUMETRIC MODELS →

STAGE

TILING

The fifteen volumetric models interlock and aggregate.

15 VOLUMETRIC MODELS
↓
INTERLOCKING
↓
AGGREGATION

> In construction